

SOLUTION

INDUSTRY SCENARIO

A healthcare company has collected patient information such as age, blood pressure, cholesterol, glucose level, BMI, and family history. The objective is to develop a machine-learning system for early diabetes diagnosis and a reinforcement-learning agent for personalised treatment recommendations.

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

1(a) Inductive Learning Approach

Inductive learning learns a general rule or model from previously observed patient examples and uses it to predict the outcome for new patients.

For this problem, a **supervised classification approach** is suitable because the dataset contains patient features and a known diabetes outcome.

A **Decision Tree classifier** is appropriate because:

- It can handle medical features effectively.
- It is easy to interpret.
- Its decision rules can be explained to healthcare professionals.
- It can identify important predictors.
- It can classify patients as diabetic or non-diabetic.

Target Variable

The target variable is:

Diabetes Outcome

- 0 → Non-diabetic
- 1 → Diabetic

Important Features

Three or more important features can include:

1. **Glucose Level** – High glucose is strongly associated with diabetes.
2. **BMI** – Higher BMI can indicate increased diabetes risk.
3. **Age** – Diabetes risk generally increases with age.

4. **Blood Pressure** – Abnormal blood pressure can be associated with metabolic risk.
5. **Family History** – A family history of diabetes can increase risk.

1(b) Handling Class Imbalance

The dataset contains more non-diabetic patients than diabetic patients. Therefore, a classifier may become biased toward the majority class.

Suitable Strategy: SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) can be used to increase the number of diabetic training examples.

SMOTE creates synthetic minority-class observations rather than simply duplicating existing diabetic records.

Advantages

- Improves representation of diabetic cases.
- Helps the model learn minority-class patterns.
- Can improve diabetic-case recall.
- Reduces the risk of ignoring diabetic patients.

SMOTE should be applied **only to the training set**, after splitting the data. Applying it before the train-test split can cause data leakage.

Another suitable method is **class weighting**:

```
class_weight = "balanced"
```

Class weights are particularly useful for a Decision Tree when false negatives are clinically important.

1(c) Data Splitting and Preprocessing

The dataset is divided using a **70:30 train-test split**.

- **70%** → Training data
- **30%** → Testing data

The training set is used to learn the model, while the test set is kept unseen and used to evaluate generalisation.

A **stratified split** should be used so that the proportion of diabetic and non-diabetic cases remains approximately the same in both sets.

Preprocessing

1. Handle missing values.
2. Identify and treat abnormal/outlier values.
3. Encode categorical variables such as family history.
4. Normalise numerical features when required.
5. Split the data into training and testing sets.
6. Apply SMOTE only to the training data.

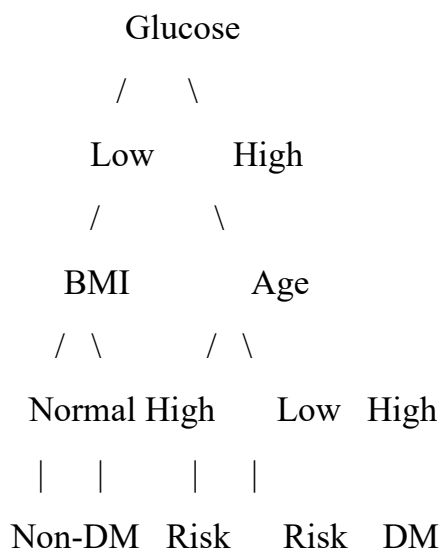
Normalisation is particularly important for models such as Logistic Regression and Naive Bayes when feature scales differ substantially. Decision Trees generally do not require feature scaling.

TASK 2: DECISION TREE FOR DIAGNOSIS

2(a) Decision Tree Classifier

A Decision Tree recursively divides the dataset according to feature values.

For example:



The actual first three levels are obtained from the trained dataset.

Root Selection

The root node can be selected using **Gini Index** or **Information Gain**.

Gini Index

The Gini impurity is:

$$[\text{Gini} = 1 - \sum_{i=1}^n p_i^2]$$

where (p_i) is the proportion of observations belonging to class (i).

A feature producing the **lowest weighted Gini impurity** provides the better split.

Information Gain

$$[\text{IG} = H(\text{parent}) - \sum_v \frac{|S_v|}{|S|} H(S_v)]$$

where (H) represents entropy.

The feature producing the **highest Information Gain** is preferred.

In a diabetes dataset, **glucose level may become the root feature** because it often provides strong separation between diabetic and non-diabetic cases.

However, the actual root must be determined from the given dataset.

2(b) Model Evaluation

The Decision Tree should be evaluated using:

Accuracy

$$[\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}]$$

Precision

$$[\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}]$$

Recall

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1-Score

$$\text{F1} = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

False Negatives

A **False Negative (FN)** occurs when a patient who actually has diabetes is predicted as non-diabetic.

In healthcare, false negatives are especially important because a missed diabetic patient may not receive early diagnosis or treatment.

Therefore, the model should prioritise **high recall/sensitivity** for the diabetic class.

Possible approaches to reduce false negatives include:

- SMOTE
- Class weights
- Adjusting the classification threshold
- Feature engineering
- Hyperparameter tuning
- Using recall-sensitive evaluation

2(c) Post-Pruning

An unrestricted Decision Tree can become very deep and overfit the training data.

Cost-complexity pruning reduces unnecessary branches.

The pruning objective is:

$$R_{\alpha}(T) = R(T) + \alpha |T|$$

where:

- $(R(T))$ = tree error
- $(|T|)$ = number of terminal nodes
- (α) = complexity parameter

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Logistic Regression

Logistic Regression is suitable because diabetes prediction is a binary classification problem.

The model estimates:

$$P(Y=1|X)=\frac{1}{1+e^{-z}}$$

where:

$$z=\beta_0+\beta_1X_1+\beta_2X_2+\dots+\beta_nX_n$$

The output is a probability between 0 and 1.

For example:

Probability = 0.82

This indicates a high estimated probability of diabetes.

Performance Comparison

The Logistic Regression model should be compared with the Decision Tree using:

- Accuracy
- AUC-ROC

AUC-ROC measures how effectively the model separates diabetic and non-diabetic patients over different classification thresholds.

When is Logistic Regression preferable?

Logistic Regression is preferable when:

- Interpretability is important.

- Relationships between predictors and log-odds are reasonably suitable.
- Probability estimates are required.
- A simpler statistical model is preferred.
- Healthcare professionals need to understand the direction and magnitude of predictor effects.

The Decision Tree is preferable when nonlinear relationships and rule-based decision boundaries are important.

3(b) Feature Importance

Decision Tree

Decision Tree feature importance is calculated from the reduction in impurity produced by each feature.

The top three features may be:

1. Glucose Level
2. BMI
3. Age

The exact ranking must be obtained from the trained model.

Logistic Regression

For Logistic Regression, feature importance can be examined using the absolute value of standardised coefficients.

For example:

Glucose Level → strong positive coefficient

BMI → positive coefficient

Age → positive coefficient

If both models identify glucose, BMI, and age among their top predictors, this provides stronger evidence that these variables are important.

If the models disagree, it may be because they capture relationships differently. A Decision Tree can model nonlinear thresholds, whereas Logistic Regression assumes a linear relationship between predictors and the log-odds of the outcome.

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

4(a) MDP Formulation

Treatment recommendation can be represented as a **Markov Decision Process (MDP)**.

An MDP consists of:

[
MDP=(S,A,P,R,\gamma)
]

States

States represent patient health stages.

Example:

S0 = Low Risk

S1 = Moderate Risk

S2 = High Risk

S3 = Controlled

Actions

The agent can choose:

A0 = Diet

A1 = Exercise

A2 = Medication

A3 = Monitor

Reward

The reward represents improvement in patient health.

Example:

+10 → Significant health improvement

+5 → Moderate improvement

+1 → Small improvement

0 → No significant change

-5 → Health deterioration

Transition Dynamics

After an action is selected, the patient's health state can change.

Example:

High Risk + Exercise → Moderate Risk

Moderate Risk + Diet → Controlled

High Risk + Monitor → High Risk

In a real medical system, transition probabilities must be learned from validated clinical evidence rather than invented arbitrarily.

4(b) Q-Learning

Q-Learning learns the expected long-term value of performing an action in a particular state.

The update equation is:

$$[Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]]$$

where:

- $Q(s,a)$ = current Q-value
- (α) = learning rate
- (r) = reward
- (γ) = discount factor
- (s') = next state

Example

Suppose:

Learning rate $\alpha = 0.5$

Discount factor $\gamma = 0.9$

Initial Q-value = 0

Reward = 5

Maximum next Q-value = 2

Then:

$$[Q_{\text{new}} = 0 + 0.5[5 + (0.9)(2) - 0]$$

$$[Q_{\text{new}} = 3.4]$$

Therefore, the updated Q-value is:

$$[Q(s,a) = 3.4]$$

These values are **illustrative**; actual Q-values depend on the patient-transition data and reward design.

Convergence Criterion

Training can be considered converged when:

$$[\max |Q_{\text{new}} - Q_{\text{old}}| < \epsilon]$$

for a predefined small threshold, such as:

$$[\epsilon = 0.001]$$

over sufficiently stable training episodes.

4(c) Final Learned Policy

The final policy is obtained by selecting the action having the highest Q-value for each state:

$$[\pi(s) = \arg\max_a Q(s,a)]$$

Example:

Patient State	Best Action
----------------------	--------------------

Low Risk	Diet
----------	------

Moderate Risk	Exercise
---------------	----------

High Risk	Medication
-----------	------------

Controlled	Monitor
------------	---------

This policy is only an **illustration**. A real clinical policy must be based on validated medical evidence and approved clinical protocols.

Ethical Considerations

1. Patient Safety

The AI system should not independently prescribe or change medication without appropriate clinical oversight.

2. Bias

Training data may contain demographic or sampling biases. The model should therefore be evaluated across relevant patient groups.

3. Explainability

Healthcare professionals should be able to understand why a prediction or recommendation was produced.

4. Human Oversight

The AI should support clinicians rather than replace clinical judgement.

5. Privacy

Patient data must be protected using appropriate access control, encryption, anonymisation/pseudonymisation, and secure data storage.

6. Continuous Monitoring

After deployment, the model should be monitored for performance degradation, data drift, bias, and unexpected recommendations.

Overall Conclusion

The proposed system combines supervised machine learning, statistical learning, and reinforcement learning for healthcare decision support. A Decision Tree can provide interpretable diabetes classification, while Logistic Regression can provide probabilistic risk estimation and statistically interpretable predictors. Class imbalance can be handled using SMOTE or class weighting, and pruning can reduce Decision Tree overfitting.

Q-Learning can model treatment recommendation as an MDP and learn policies from patient-state transitions. However, because treatment recommendations directly affect patient safety, the reinforcement-learning component should be used as a clinical decision-support system with human supervision, not as an autonomous medical decision-maker.